Foundation for Virtual and Augmented Reality Systems

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Week – 02

Lecture – 09

Lec 9: Issues & Challenges

Hello and welcome to the NPTEL MOOCs course on Foundations for Virtual and Augmented Reality Systems. We are continuing our discussion on the issues and challenges in this lecture number 5. So, earlier we talked about a few issues in VR, namely the technological issues and the psychological issues. Technological issues refer to immersion. Psychological issues refer to the concept of presence.

What these issues broadly refer to is that through technology we can generate sensory stimuli, and our psychology helps us to perceive those stimuli. Now there should be a match between them so that we are able to perceive the virtual world as the real world, without being able to distinguish between the two worlds. The closer we are to this goal, that is the goal of any VR system, the closer we are to this goal, the better immersive experience we are providing to the user. But there are other issues as well in VR systems.

Let us have a quick look at those issues. The goal of any VR system is to create a high degree of presence or, in other words, to make people believe they are in the real world while actually they are in the virtual world. But there is a very interesting thing here. When I say that we are trying to make people feel that they are in the real world, while they are actually in the virtual world, what I am referring to is the term 'real'.

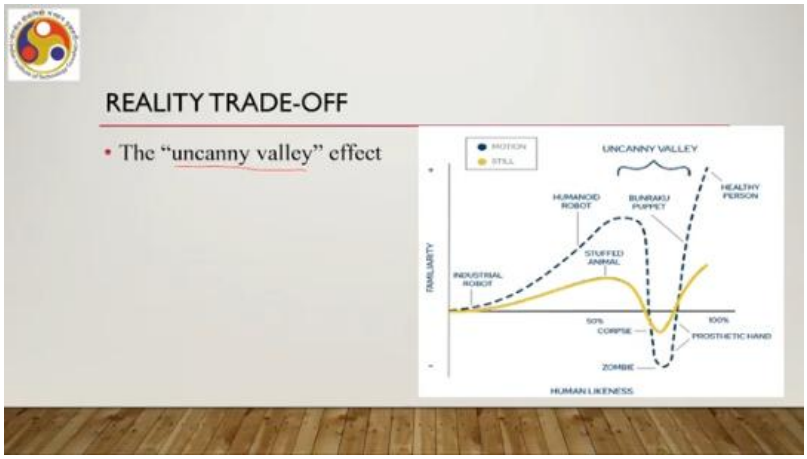
Now, that is not as easy as it sounds. The answer to this question may not be as easy as it sounds. For some people, reality is the gold standard. That is, whatever we perceive, whatever we sense in the real world, that is reality. So for those people, if you are making a system, then we should try to match the experience as close to our real world experience as possible.

That is, we can say, the goal of a VR system for that group of people is as close to reality as possible. That is not the end of the story. For some people, when we are building a virtual reality system, the idea of reality should be something beyond our lived reality experience. That means, whatever we are experiencing in our day-to-day life, in our so-called real world, should not be exactly mimicked in the virtual reality experience. So, there the concept of reality should be

extended beyond our lived perception of or lived experience of reality. So, others, those people consider reality in virtual reality to be something that surpasses our idea of reality. So, that should be the goal of a virtual reality system for that group of people. Then, as a developer, if you are given the, if you are asked to build a system, how do you decide? It is not easy, not easy to decide.

What some consider a good VR system, because it matches their sense of reality, may not be considered a good VR system to some other group of people, because it does not match their sense of or their expectation of reality. Now, if you are surpassing reality, then that reality may be good for some people, but for others it may seem creepy or difficult to comprehend. So, there is some confusion in terms of what exactly reality is. There is something called the uncanny valley effect.

Now, this actually is an outcome of a scientific study which showed how a human user is likely to perceive an artificial object and how closely the user is likely to feel the object is realistic. What does this effect suggest? Digital representations of humans appearing almost real evoke feelings of unease or eeriness rather than familiarity due to subtle imperfections. Although here we are using the term digital representations, originally the study was done with respect to robots, but that idea can be extended to the digital world.



So, essentially it says that no matter how hard we try to make something appear like a human or something appear realistic, there will always be some sort of feeling of unease or eeriness due to very subtle imperfections that might be there which are not easily recognizable or perceivable at the very outset. Now, this idea is explained in this plot, which also suggests that the effect, the uncanny valley effect, is visible when the human likeness is reaching about between 70 to 85 percent of the similarity.

So this idea is illustrated in this plot. Now there are two lines as you can see: one dotted line, this one, and one solid yellow-colored line, this one. The dotted line indicates a moving object whereas the solid line indicates something which is stationary. On the x-axis there is human likeness and on the y-axis there is familiarity. Now, near the origin, we have industrial robots, which anybody can see as totally different from humans.

So, it does not matter whether they are perceived to be familiar or not, they are totally unfamiliar. So, that is one extreme. Now, as we move along the human likeness axis, Essentially, this indicates that we are trying to make the artificial object, or in this case robot, or in the context of VR, it may be digital objects. As we are trying to make it more and more human-like, the familiarity effect is also increasing for both objects in motion and stationary objects.

In both cases, we can see that it is increasing. up to a certain extent. In both cases the increase happened up to a certain extent, then we can see that the familiarity started decreasing. So, for stationary objects it is decreasing like this, whereas for objects in motion it is decreasing more sharply. Of course, it is another way of telling that when the object in motion loses familiarity, that is pretty sharp as compared to when the object is stationary.

Now, this decrease goes on for some time although we are trying to increase the familiarity, increase the human likeness here along this axis, as you can see we are moving towards the positive direction. However, the familiarity keeps on decreasing and reaches a minimal point, and then again starts increasing, and after some time it keeps on increasing. Now, this region where the Although human likeness is increased, familiarity started showing signs of decrease and the point where it started rising again, this is called the uncanny valley.

This is the region where although we are assuming that we have made our object appear like a human, but in reality what is happening is that due to subtle imperfections as mentioned here, the perception of that object to the user may be creating some feeling of unease or eeriness. That is somewhat counterintuitive as you can see. And this region is

typically when the human likeness reaches between 70 to 85 percent. So, this region is the uncanny valley region.

where the likeness we have increased to a significant extent between 70 to 85 percent, but familiarity decreased. So, that is the uncanny valley effect and it has its implications when we are trying to create artificial environments in virtual reality. Although we may assume that we are creating very realistic scenarios with realistic objects, but that realism may not be appearing to be realistic to the end user that is the significance of this uncanny valley effect. So, here what it implies is that we are having a reality that is more we are increasing our reality that is more and more human-like figure, but we are not getting the desired perceptual effect.

So while creating a virtual reality experience we should keep in mind that there may be an uncanny valley effect, that is we are trying to exactly match reality but that match may not always lead to a good experience if it falls within this uncanny valley zone then this our experience may start deteriorating. So, there is this trade-off how much reality should be good reality or up to how much we should go to give a good experience. Now, one thing is there: that presence does not require photorealism.

Other cues may be equally useful. So, we do not need to always create, say in the context of our earlier discussion, we do not need to always create a human-like object. There can be other cues which may help us achieve higher presence. So, in order to do that, a VR system developer or VR content creator should consider a few things. One is representational fidelity.

One is interaction fidelity and the third is experiential fidelity. So, what does representational fidelity say or refer to? The degree to which the VR experience conveys a place that is or could be on Earth. The familiarity with a place on earth, the degree to which the VR experience matches with that familiarity or exploits that familiarity, that is representational fidelity.

So, at the high end very good photorealistic immersive film as content can refer to high representational fidelity that is we are using 360-degree video or images of the real world in our virtual environment. then that refers to the highest representational fidelity. On the other hand, if we are using purely abstract and non-objective worlds, creating such worlds which are totally unfamiliar to us in our real-world experience, then that refers to extremely low or the other end of the representational fidelity spectrum.

Then we have interaction fidelity. The degree to which physical actions for a virtual task correspond to physical actions for an equivalent real-world task. This again tries to exploit our real world experience. So, at the high end or one end of the spectrum, we have intense realistic physical actions for virtual tasks.

So, we are doing physical activities in the real world to translate them to virtual activities in the virtual world. At the other end, no physical motion at all for virtual tasks. Say for example, we are sitting, but in the virtual world, we are moving forward. by button press. So that refers to very low interaction fidelity.

Then we have experiential fidelity. The degree to which the user's personal experience matches with the intended experience of the VR creator. So a free roaming world with endless possibilities. And every usage leading to a different experience has low experiential fidelity because that is not what happens in the real world. Whereas, real world constraints put into the virtual world navigation, movement, and interaction means more experiential fidelity.

So, the closer the experience in the virtual world is to the real world, the higher is the experiential fidelity. So that is what we should consider along with immersion and presence. We should keep in mind the different fidelities: representational, experiential, and interaction fidelities. And why keep those things in mind? Because of this uncanny valley effect, we should be aware of this effect and we should ensure that these fidelities are maintained to the highest degree possible so that

Such effects do not affect our virtual world experience. So, with that we close our discussion on VR issues and challenges. Let us move to the next part that is AR issues and challenges. Now, in AR as we have seen earlier, there is not much immersion. Immersion is not a major concern.

So, psychological effect is less important here as compared to VR. So, accordingly the challenges are mostly technological rather than psychological. So, let us revisit the definition of AR to understand what are the challenges. This definition was given by Azuma in 1997. So, there are three requirements for any system to be considered as an AR system.

One is it should combine real and virtual worlds. So, virtual content should be overlaid on the real world representation. It should be interactive in 3D in real time, it should be

interactive in real time and it should be registered in 3D. So, from these characteristics we can say what are the or we can get some idea of what are the challenges in AR systems.

So, the first thing is the combination of real and virtual content. So, we are getting a representation of the real world on our device and we need to overlay digital content or virtual content on this real world view. In order to do that, we need suitable display technology. That is our first issue. An appropriate display technology.

Of course, this is there in VR as well, but in AR also this issue is there. Interactive in real time refers to the fact that we should have or we should support suitable forms of interaction so that the user should be able to interact with the digital content overlaid on the real world representation. This refers to different input devices as well as different input modalities. And finally, registered in 3D.

It relates to the ability of a system to anchor virtual content in the real world such that it appears to be a part of the physical environment. So, as we have discussed earlier, in simple words, it means that whenever we are augmenting virtual content in the real world representation, it should appear at a suitable place rather than appear randomly so that it mixes seamlessly with the real world objects. Now, to register virtual content in the real world the pose of the viewer with respect to some anchor in the real world must be determined.

Now, pose refers to two things: position and orientation in our context. So, unless we are able to do this proper estimation of pose or proper estimation of position and orientation with respect to some real world anchor in the virtual world. with respect to some real world anchor, it will be difficult or impossible to register the content in 3D. Now, this pose determination has one or two phases. There are actually two conceptually different things.

Both can be done in one phase or we can have separate phases for each of those. One is the registration phase. It determines the pose of a viewer with respect to the real world anchor. That is the registration phase. The other one is the tracking phase.

So first we have registration, then we have the tracking phase. That is why a one-time determination of pose is not sufficient as the user is likely to move with an AR application. We need to update the pose of the viewer relative to a previously known pose. So, essentially, the change of viewpoint we should update with respect to pose, so that the objects appear properly; the mixing of digital and virtual content, digital and real world content, appears properly and suitably to the user.

So, to summarize, we have two things, two stages rather. One is the registration phase. At the beginning, we do it when the user starts using the system, and then as time progresses, as the user movements occur, we update the pose with respect to the current pose. So, these are the issues in AR. So, let us summarize what we have learned.

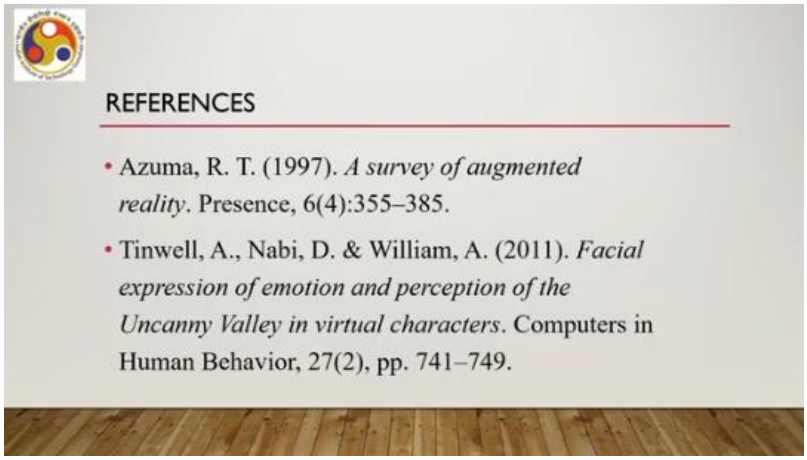
So, we started our discussion with VR issues. There we talked about two major concerns. One is technological, one is psychological, when we are talking about VR systems. And these led to two major concepts. One is immersion, which is tied to the technological aspect, the technological issues of VR.

The other one is presence, which is tied to the psychological issues of VR. While discussing these issues, we also noted that there is something called the uncanny valley effect. Now, this effect tells us that although our objective may be to make environments as familiar to the user as possible with respect to their real world experience. Sometimes this familiarity may lead to some effects, particularly uneasiness or creepiness in the mind of the user.

So, we should be aware of this effect and try to avoid that while building systems, VR systems. And in order to do that, we should be aware of a few things namely representational fidelity, experiential fidelity, and interaction fidelity. For AR, the issues are mostly technological, related to proper display technology, proper input mechanisms and technology, as well as estimation of pose at the beginning and during the usage of the system.

So, that pose estimation and updating is popularly called tracking, tracking of the pose. So, the tracking technology, input technology, and display technology are major concerns, major issues in AR. In the next lecture, we are going to take up another major concern particularly related to VR systems, that is cyber sickness. So, in this lecture, we have talked about a few issues, major issues. One more issue we are going to take up in the next lecture is cyber sickness, which has three parts, or discussion will be carried out in three parts.

The causes of cyber sickness, the measurement, how we can measure cyber sickness, and the mitigation strategies, how we can reduce or eliminate cyber sickness while letting the user experience virtual reality. So, these are the references that are used. In this part of the lecture, you can go through them. They are good reads.



With that, I would like to end this lecture here. I hope you have enjoyed and understood the concepts clearly. See you in the next lecture. Until then, goodbye and thank you. Happy learning.